LESSON PLAN

ADVANCED AIRCRAFT TECHNOLOGY (AAT) (31)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	01
Total No. of Lesson Plans	12	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	General safety precautions to be observed while working with power machines and power tools.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	General safety precautions to be observed while working with power machines and power tools. Introduction to Lathe, Milling, Grinding, Shaping and Radial drilling machine
5. Specific Objectives	31.1.1.1 Trainee will be able to perform various general engineering works

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of safety gear usage and machine startup checks under instructor supervision.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary purpose of wearing safety goggles?
(b) Why must loose clothing be avoided near rotating machinery?
(c) What is the first step before operating any power tool?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• General safety precautions for power machines • Introduction to Lathe machine components • Introduction to Milling machine components • Introduction to Grinding machine components • Introduction to Shaping machine components	L & Ds	PPT, IB	15 min
(b)	• Specific safety rules for Lathe operation • Specific safety rules for Milling operation • Specific safety rules for Grinding operation • Specific safety rules for Shaping operation • Specific safety rules for Radial drilling operation	L & Ds	PPT, PDF	20 min
(c)	• Emergency stop procedures for all machines • Proper handling of cutting tools and workpieces • Maintenance and inspection of power tools • Storage and disposal of swarf and chips • Reporting of machine defects or hazards	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) List three critical safety precautions specific to the Lathe machine.
(b) Explain the correct procedure for changing a grinding wheel.

12. Home Assignment and Follow Up	Prepare a brief note on the specific safety precautions required for the Lathe machine to be discussed in the next lesson on turning and facing operations.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Procedure of turning, facing, thread cuttings, grooving, knurling and tapering in lathe machine.	Refer to the master syllabus reference list	LP 02

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min